*Equilux*

# Pay, proven equal.
# No salary revealed.

The first employee-verifiable pay-transparency reporting protocol. A company proves its legally required gender pay gap report was computed from its complete payroll — every salary the payroll provider's own figure, confirmed by the employee it belongs to — while every salary stays in Midnight's private state.

Midnight Buildathon · Wave 2 | Compact · TypeScript · Apache 2.0 | github.com/AustinChris1/Equilux

# The law now demands a number companies cannot prove.

**7 June 2027**

FIRST REPORTS DUE · 2026 PAY DATA

**250+**

EMPLOYEES, ANNUAL · 150+ SAME DATE, TRIENNIAL

**≥ 5%**

UNJUSTIFIED GAP IN A CATEGORY → JOINT PAY ASSESSMENT

Article 9 requires seven indicators. Article 7 gives every employee the right to ask what their category earns. And today every one of those reports is **self-declared**: nobody can verify the numbers without seeing everyone's salary.

THE PROBLEM

# Trust me, or prove it.

## Today

HR exports payroll to a spreadsheet, computes a gap, publishes a PDF. An employee cannot check they were counted. A works council cannot check nobody was dropped. A regulator can only audit by subpoenaing the whole payroll — the data the law is trying to protect.

## With Equilux

The payroll provider commits every payroll row, hashed. The works council opens every row before anyone enrolls. Each employee seals their own record and proves it matches an unclaimed row. The circuits then check all seven Article 9 figures. Only the figures and the proof go public.

Equilux does not replace the HRIS. Payroll stays in Personio, DATEV or Workday. Equilux is the **notary on top**: it proves the set was complete and the published figure matches it.

# The works council opens the payroll. All seven Article 9 figures are proven.

| | WAVE 1 | WAVE 2 |
|---|---|---|
| Who checks the payroll | the employer, on its own data | **the provider commits each row; the works council opens every one** |
| A salary that is not payroll's | the employer could vouch for it | **rejected** — the record must open an unclaimed payroll row |
| Figures proven | the mean gap | **all seven Article 9 indicators**, including variable pay and quartiles |
| Small groups | designed | **published only at k = 3**; smaller groups stay suppressed |
| Where the proof ran | a local Midnight node | **Midnight preprod**, blocks 2,787,450–2,787,556 |
| What a judge can open | the site, after installing Docker | **/verify** reads the chain; 11 cheats on the public site, no install |
| Tests on the compiled circuits | 9 | **45** (55 tests in all) |

— WHAT THE CIRCUIT PROVES

# Trust neither side. Verify both.

## Completeness

The payroll provider commits every payroll row before anyone enrolls; the report must cover exactly those rows. Leaving someone out — even by never enrolling them — is rejected.

## Every salary is payroll's

A record counts only if it opens to an unclaimed payroll row. The employer cannot invent a person or change anyone's pay; an employee cannot inflate their own.

## No small-group leaks

A category's pay is published only when both groups have at least three people. Smaller groups show headcounts only.

## Exact figures

All seven Article 9 indicators are claims the circuit recomputes: mean, median, category gaps, variable-pay gaps, who receives variable pay, and quartile bands. One basis point off fails.

Plus an **inclusion receipt** for every employee: a Merkle path proving their record is inside the numbers, revealing nothing about anyone.

# The council opens every row. Then the employee. Then both reports.

`01 · declareRoster`

## Commit

The payroll provider commits a hiding hash of every row, in one transaction.

`02 · confirmPayroll`

## Council

The works council opens every committed row. Enrollment stays closed until it does.

`03 · enroll`

## Bind

The employee seals their record and proves it opens an unclaimed payroll row.

`04 · publishReport`

## Basic pay

Mean gap, median gap, and each category's gap, suppressed below k = 3.

`05 · publishVariablePay`

## The rest

Variable-pay gaps, who receives it, and the gender mix of each pay quartile.

`06 · checkReceipt`

## Receipt

An employee proves their record is in the tree, against the on-chain root.

# Private state holds the payroll. Public state holds the proof.

## Private state · witnesses

- salary, variable pay, gender marker, category
- employee secret and the payslip nonce
- employer, provider, and works-council keys
- the full 16-slot payroll witness
- every claimed figure, including each quartile band

## Public ledger

- commitments — a depth-5 Merkle tree
- payroll-row hashes, nullifiers, bound commitments
- three role keys, headcount, council confirmation
- both reports: all seven Article 9 indicators
- the zero-knowledge proof for every transition

Every witness that reaches the ledger passes `disclose()`. The compiler refuses anything else.

— WAVE 2 EVIDENCE · REAL PROOFS ON A MIDNIGHT NETWORK

# Fourteen people. Two reports. On public preprod.

**6**

PROVING CIRCUITS · PROVER + VERIFIER KEYS

**45**

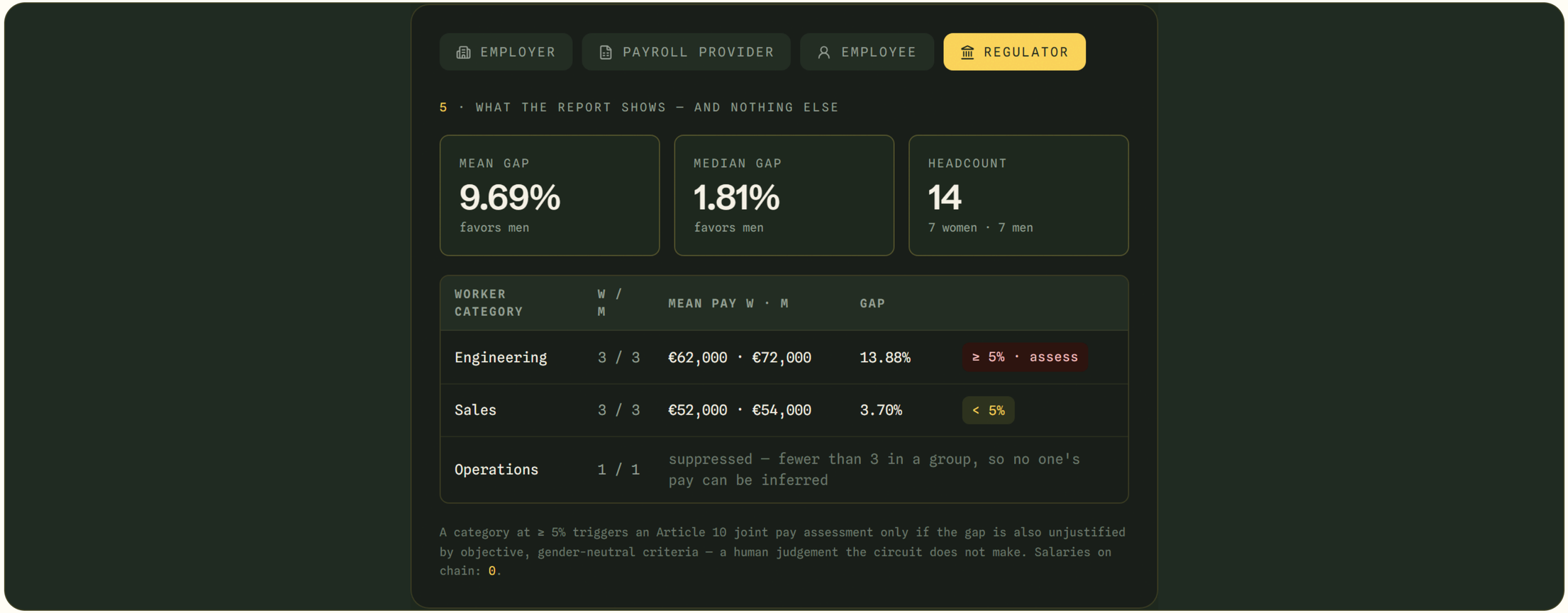
OF 55 TESTS · ON THE COMPILED CIRCUITS

**0**

SALARIES ON CHAIN · EVER

```
preprod contract 598b346e…d1cb
declareRoster · 14 payroll-row hashes                    (block 2,787,454)
confirmPayroll · works council opens every row           (block 2,787,463)
14 enrollments, each bound to its own row                (blocks 2,787,468–2,787,529)
publishReport proven · mean 9.69% · median 1.81%         (block 2,787,543)
publishVariablePay proven · variable-pay mean gap 27.50% (block 2,787,556)
Engineering 13.88% flagged · Sales 3.70% · Operations suppressed · same figures as /verify
```

# No Docker. No install. The real circuits, on the public URL.



# Honest about what a proof can prove.

| IT DOES NOT CLAIM | WHAT IS TRUE INSTEAD |
|---|---|
| Salaries hidden from HR | HR still holds payroll. Hidden from the public, colleagues, the regulator and Equilux — and the employer can no longer supply its own figures. |
| An honest payroll by itself | The works council must open every row. A provider, employer and council who collude could still commit false rows. Each employee sees their own. |
| Judging what is justified | Category gaps ≥ 5% are proven and flagged. Whether a gap is justified is a human judgement. |
| A finished legal filing | All seven Article 9 indicators are proven. The instance is 16 records and 4 categories, not a 250-person employer. |
| Any visitor can write on preprod | The 14-person run is on preprod. Connecting Lace is enough to read; a write from that wallet needs tDUST. |
| A model of identity | Binary gender markers, as the Directive's reporting annex is written today. |

THREE WAVES

# Built in the open, wave by wave.

Wave 1 · delivered

## Prove it works

Contract with private state · proven end to end on a Midnight network · employer / employee / regulator workspace · honest scope.

Wave 2 · this wave

## The product

Works council · payroll-row binding · all seven Article 9 indicators · the 14-person flow on preprod · /verify and the filing pack · Connect wallet · Personio / DATEV import.

Wave 3

## A named pilot

One EU employer and its works council, on their own payroll export. A real Personio sandbox. The first full write from a visitor's wallet.

WHO BUYS · WHAT WE ASK

# The notary for Directive 2023/970.

## Buyers

Works councils that don't trust the numbers. GDPR-sensitive employers. Regulators who want completeness without holding payroll.

## A plugin, not a rival

Workday, Personio, DATEV compute pay. Equilux imports their export, the provider commits its rows, and the report is proven — next to them, not instead of them.

## Why Midnight

A forced legal deadline plus a problem only dual-ledger selective disclosure solves honestly. Incumbents can copy a dashboard; they cannot copy the proof.

Equilux

youtu.be/AmFv_UKSj4A
github.com/AustinChris1/Equilux
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